As of: March 21, 2025
Received: March 14, 2025
Status: [redacted]
Tracking No. m89-7h7t-lnqp
Comments Due: March 15, 2025
Submission Type: API

PUBLIC SUBMISSION

Docket: NSF_FRDOC_0001
Recently Posted NSF Rules and Notices.

Comment On: NSF_FRDOC_0001-3479
Request for Information: Development of an Artificial Intelligence Action Plan

Document: NSF_FRDOC_0001-DRAFT-1430
Comment on FR Doc # 2025-02305

Submitter Information

Email: [redacted]
Organization: Nuclear Energy Institute

General Comment

See attached file(s)

Attachments

NEI Comments AI Action Plan_FINAL

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March 14, 2025

Faisal D'Souza
National Coordination Office (NCO)
2415 Eisenhower Avenue
Alexandria, VA 22314

Subject: NEI Comments for Development of Federal Artificial Intelligence Action Plan; Docket NSF_FRDOC_0001-3479

Dear Mr. D'Souza,

On behalf of the Nuclear Energy Institute's (NEI)[1] members, we are pleased to offer comments regarding development of the federal Artificial Intelligence (AI) Action Plan, as outlined in President Trump's signed Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence), on January 23, 2025.

The recent AI boom has resulted in surging demand for reliable electricity, primarily due to the rapid deployment of data centers and the need for more advanced computational power, which consumes significant amounts of energy. In the United States, data center demand is expected to account for 12% of total power consumption by 2030 (a 3x increase from current levels). Meanwhile, large hyperscalers and other data center developers have placed an emphasis on increasingly clean power for their operations with aggressive targets.

Nuclear power is uniquely situated to meet this moment as the nation's largest source of clean, reliable power.

Our nation's 94 operating nuclear units play a vital role in ensuring a safe, reliable, and affordable electric grid to serve customers across the United States. Collectively, our fleet is pursuing 3 GW of power uprates, and 3 GW of historic plant re-starts, the equivalent of adding approximately six large light-water-reactors to the grid, which would power over 5 million homes. In addition, the majority of the operating fleet is planning for life extensions out to 80 years, as demand from large energy consumers and customers skyrockets.

[1] The Nuclear Energy Institute (NEI) is responsible for establishing unified policy on behalf of its members relating to matters affecting the nuclear energy industry, including the regulatory aspects of generic operational and technical issues. NEI's members include entities licensed to operate commercial nuclear power plants in the United States, nuclear plant designers, major architect and engineering firms, fuel cycle facilities, nuclear materials licensees, and other organizations involved in the nuclear energy industry.

In the last year, several U.S. tech companies have signed contracts for advanced reactors to support their significant data center growth, with deployment plans as early as 2030. In fact, new nuclear partnerships with tech customers totals nearly 30 GWe of commitments. In addition, some tech companies are also turning to existing nuclear assets to help power their operations through Power Purchase Agreements. In some cases, large energy consumers are forging new business models by partnering together to aggregate their demand for advanced nuclear technologies. Clearly, there are different approaches to integrating nuclear power, but the commonality is that nuclear power is being valued for its 24/7, clean, reliable attributes by large electricity customers.

President Trump has made clear that global AI dominance is a national security imperative and priority. This will be achieved by nations that can secure the largest volumes of reliable power, with speed to market, paired with a robust private sector and dominant manufacturing capabilities. Nuclear power can uniquely meet this demand, while also contributing to mutual goals of economic competitiveness, energy dominance, and domestic job creation. The nuclear energy industry is seeing a significant resurgence as shown by strong bi-partisan support, the passage of key legislation at the federal and state levels, and opportunities for new partnership. Such partnership will extend beyond technological advancements and allow for broader global strategic influence in the global AI race against Russia and China. Further, recent geopolitical conflicts have forced nations to see nuclear with new eyes, and several dozen countries (including the United States) pledged to triple nuclear power by 2050. The time to honor that pledge is now.

The signal to the marketplace is clear: demand for nuclear is strong and growing. As officials across the interagency begin to develop the AI Action Plan as a matter of national priority, we urge you to consider the fundamental role that nuclear power will play, in meeting the power demands to build out the nation's technological capacity and leadership.

Thank you for your consideration of our comments. If you have any questions or would like to discuss further, please contact me, or Hilary Lane ([redacted]

Sincerely,

John Kotek
Senior Vice President

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